

Responses to the Review Comments of BDR-D-25-00303R3

Original Manuscript ID: BDR-D-25-00303R3

Original Article Title: “Dual Balanced Large Marginal Machine for Imbalanced Data Classification”

To: Big Data Research Editor

Re: Response to reviewers

Dear Editor,

Thank you for allowing a resubmission of our manuscript, and we particularly appreciate the Editor-in-Chief for granting us the opportunity to conduct a round of revisions to address the reviewers’ comments.

We are uploading (a) our point-by-point response to the comments (below) (response to reviewers, under “Author’s Response Files”), (b) an updated manuscript with orange highlighting indicating changes, and (c) an Supplementary file submitted in accordance with the reviewers’ requirements.

Best wishes,

Pingping Gu, Denghao Dong, Quan Shi, Fudi Wang and Guojun Huang*

The response letter is structured as follows:

- Response for Editor: Page 2.
- Response for Reviewer 7: Page 3.
- Response for Reviewer 8: Page 4-8.

EDITORGeneral Comments:

I have completed my evaluation of your manuscript. The reviewers recommend reconsideration of your manuscript following minor revision and modification. I invite you to resubmit your manuscript after addressing the comments below. Please resubmit your revised manuscript by 07/05/2026. When revising your manuscript, please consider all issues mentioned in the reviewers' comments carefully: please outline every change made in response to their comments and provide suitable rebuttals for any comments not addressed. Please note that your revised submission may need to be re-reviewed.

REVIEWER 7

General Comments: The manuscript has undergone three rounds of revision and is now highly polished. The authors have addressed the reviewers' concerns regarding comparison with the latest state-of-the-art models, scalability for Big Data paradigms, experimental reproducibility, and terminology consistency. The paper provides a meaningful incremental advance in margin distribution learning. While the $O(n^3)$ complexity is a hurdle for "Big Data," the provided theoretical paths toward distributed and approximated versions justify its publication.

Reviewer 7, Final Suggestion: Before final publication, ensure that the "Supplementary Material" links are finalized and that all magenta/orange highlighting from the revision process is removed for the final production version.

REVIEWER 8

General Comments: The authors have moved the manuscript forward in this round. The new Remark 2 clarifying “quantile distance” versus “composite distance” is useful, the MLP+Focal configuration is now properly specified, and the limitations discussion in the conclusion is a real improvement. That said, the new head-to-head against DeepSMOTE, LDAM, and Balanced Softmax introduces problems that need to be resolved, and several issues that should have been caught in earlier rounds are still present. I recommend major revision.

Reviewer 8, Strength 1: The new Remark 2 clearly distinguishes the two distance concepts that were previously conflated. This was a real source of confusion in R2 and the fix is well done.

Reviewer 8, Strength 2: The expanded MLP+Focal configuration in Section 5.3 (architecture, optimizer, β values, batch size, LR schedule, grid search ranges) is detailed enough for reproducibility.

Reviewer 8, Strength 3: The limitations paragraph in the conclusion (training complexity, extreme imbalance, concept drift, unstructured data) gives a more balanced positioning of the method than earlier versions.

Reviewer 8, Strength 4: The scalability discussion (ADMM, mini-batch, Random Fourier Features) at least acknowledges the journal’s scope, even if it is not yet demonstrated.

Reviewer 8, Major Concern 1: The LDAM configuration as described is methodologically incoherent. The authors write that LDAM “adapts ResNet-18 for tabular data.” ResNet-18 is a 2D convolutional architecture designed for image inputs. There is no standard way to apply it to tabular features of dimension 3 to 61, which is the range in Table 1. The authors need to explain exactly what architecture was used, replace it with a sensible tabular backbone (MLP with LDAM loss would be the natural choice), or drop the claim. As it stands, readers cannot reproduce or evaluate this baseline.

Author response: We thank the reviewer for identifying the architectural inconsistency. We implemented LDAM with a tabular MLP backbone rather than ResNet-18: two hidden layers of 128 and 64 units with batch normalization, ReLU, and dropout 0.3. The LDAM loss uses class-dependent margins with maximum margin $\Delta = 0.5$, scale 30, and deferred class reweighting after epoch 160. DeepSMOTE and Balanced Softmax are also implemented with explicit tabular architectures, and all three baselines are evaluated by stratified 5-fold cross-validation. Thus, the revised comparison is directly applicable to feature vectors with 3–61 dimensions and is reproducible from the released implementation.

Author action:

►► Location in the revised manuscript: Section 5.3, the paragraph introducing the three imbalance-learning baselines (beginning “We additionally evaluate three recent imbalance-learning baselines...”).

Original text: “Additionally, we compare with three recent deep learning methods representing different paradigms in imbalanced learning: DeepSMOTE (Dablain et al., 2022) employs an encoder-decoder architecture with latent dimension 128, three hidden layers (256-128-64 neurons), ReLU activation, and penalized loss weight 0.1; LDAM (Cao et al., 2019) adapts ResNet-18 for tabular data with class-dependent margin parameter $\Delta = 0.5$ and deferred re-weighting schedule; Balanced Softmax (Ren et al., 2020) implements frequency-based reweighting with temperature parameter $T = 1.0$. All experiments employ 5-fold cross-validation.”

Revised text: “We additionally evaluate three recent imbalance-learning baselines using tabular architectures. DeepSMOTE uses an encoder-decoder with latent dimension 128 and encoder widths 256–128–64; minority samples are interpolated among latent-space nearest neighbors and decoded before classification, with reconstruction penalty weight 0.1. LDAM uses the same two-hidden-layer tabular MLP backbone as the other neural baselines, class-dependent margins with maximum margin $\Delta = 0.5$, scale 30, and deferred class reweighting after epoch 160. Balanced Softmax uses the same MLP backbone and adds the logarithm of the training class counts to the logits with temperature $T = 1.0$. All three baselines use Adam (learning rate 0.001, weight decay 1×10^{-5}), batch size 32, at most 200 epochs, early stopping patience 10, and stratified 5-fold cross-validation. The released code reports the seed and each fold’s Accuracy, AUC, and F1-score.”

Revised Table 7 (in the revised manuscript, Sec. 5.3): each entry reports Acc/AUC/F1-score averaged over five stratified folds, the best value in each column is bold, and the numbers that changed in this revision are shown in red.

TABLE I: Performance Comparison (Acc/AUC/F1-score) Among DBUPLDM, MLP+Focal Loss, DeepSMOTE, Tabular-MLP LDAM, and Tabular-MLP Balanced Softmax on 17 Benchmark Datasets

Model	Monk1	Monk2	Monk3	Pima	WDBC	Echo
DBUPLDM	68.29/72.13/0.69	67.82/78.64/0.76	88.89/92.43/0.89	79.25 /79.34/0.72	87.50/91.43/0.89	92.72 /83.66/0.76
MLP+Focal	92.80 /98.59/0.93	77.38 /93.28/0.78	96.03 /99.26/0.96	70.95/82.56/0.71	96.14 /99.62/0.96	77.81 /88.28/0.78
DeepSMOTE	93.71/98.85/0.94	87.38 /94.36/0.88	96.93 /99.52/0.97	74.86/82.79/0.75	96.31 /99.36/0.96	77.01 /85.42/0.78
LDAM	92.99/98.62/0.93	83.20/90.57/0.83	96.03/99.21/0.96	77.47/ 83.11 /0.77	97.36 /99.60/0.97	84.64/ 88.75 /0.85
Balanced Softmax	96.94 / 99.72 /0.97	86.69/ 96.08 /0.87	96.57/99.46/0.97	73.69/83.01/0.74	96.67 / 99.63 /0.97	82.39/87.99/0.83
Model	Iono	Habe	Stat	Germ	Aust	Vote
DBUPLDM	78.85/89.23/0.76	98.82 /85.51/0.86	92.16 /79.46/0.91	76.40 /73.21/0.78	87.24 /83.44/0.84	95.32 /87.29/0.79
MLP+Focal	94.29 /98.79/0.94	36.30/67.95/0.31	81.11/87.61/0.81	60.90/77.54/0.62	85.51/92.44/0.86	58.62/64.71/0.58
DeepSMOTE	94.86 /98.58/0.95	71.57/69.76/0.71	81.48/89.33/0.81	70.90/77.13/0.72	85.94/92.75/0.86	60.92/64.52/0.61
LDAM	93.73/98.61/0.94	71.56/66.54/0.66	80.74/87.17/0.81	75.50/77.19/0.75	84.93/92.65/0.85	63.91/64.40/0.64
Balanced Softmax	94.86 / 98.97 /0.95	68.95/69.49/0.70	84.07/ 89.44 /0.84	71.00/ 77.58 /0.72	84.93/ 92.96 /0.85	62.76/66.56/0.63
Model	Diab	Fert	Sonar	Ecoil	Plrx	Avg
DBUPLDM	82.09 /80.12/0.79	96.00 /92.88/0.88	76.85/72.42/0.78	94.49/82.18/0.75	69.51 /68.72/0.68	84.16/82.78/0.81
MLP+Focal	70.95/82.56/0.71	32.00/59.05/0.35	82.23/91.93/0.82	96.03 /98.90/0.96	38.36/44.60/0.37	73.38/83.98/0.73
DeepSMOTE	74.86/82.79/0.75	87.00/56.44/0.82	81.29/91.03/0.81	96.03 /98.81/0.96	54.37/42.26/0.55	81.50/83.75/0.81
LDAM	77.47/ 83.11 /0.77	84.00/63.30/0.83	80.78/90.76/0.81	96.03 /98.78/0.96	62.09/42.36/0.56	82.50/83.81/0.82
Balanced Softmax	73.69/83.01/0.74	36.00/56.37/0.41	84.63 / 92.22 /0.85	96.03/ 99.13 /0.96	52.67/39.17/0.51	78.97/ 84.16 /0.79

Reviewer 8, Major Concern 2: Please release the code, seeds, and fold-level outputs for Table 7. The ordering LDAM over Balanced Softmax over DeepSMOTE is very consistent across the 17 datasets and across Accuracy, AUC, and F1. This ordering is plausible on its own, but the consistency is unusual enough that the paper should provide a reproducibility package (code, random seeds, per-fold numbers). This would also address Reviewer 5’s original reproducibility request in a more complete way.

Author response: We agree that the unusually consistent ordering required auditable evidence. The reproducibility package now contains the complete baseline source code, the fixed seed (42), the exact run configuration, stratified fold assignments, and one record per dataset, method, and fold with Accuracy, AUC, F1-score, and the number of epochs trained. All Table 7 values are computed as the arithmetic mean of the five fold-level records; no fold or metric is removed or selected. The new experimental results are reported in Revised Table 7 above. The code and the benchmark datasets are openly released at https://github.com/Jlchen08/dbupldm_code, and a Data and Code Availability statement has been added to the revised manuscript. The results from the new experiments also lead to a more measured conclusion, which we state explicitly in Sec. 5.3: DBUPLDM improves on MLP+Focal in average accuracy (84.16% versus 73.38%) and average F1-score (0.81 versus 0.73), while MLP+Focal has a higher average AUC (83.98% versus 82.78%); and against the three deep baselines DBUPLDM has the highest average accuracy (84.16%), Balanced Softmax the highest average AUC (84.16%), and LDAM the highest average F1-score (0.82).

Author action:

►► Location in the revised manuscript: Section 5.3, Table 7 (Performance Comparison) and the comparison paragraphs below it; the Data and Code Availability statement; code and data at https://github.com/Jlchen08/dbupldm_code.

Original text: “Additionally, we compare with three recent deep learning methods representing different paradigms in imbalanced learning: DeepSMOTE (Dablain et al., 2022) employs an encoder-decoder architecture with latent dimension 128, three hidden layers (256-128-64 neurons), ReLU activation, and penalized loss weight 0.1; LDAM (Cao et al., 2019) adapts ResNet-18 for tabular data with class-dependent margin parameter $\Delta = 0.5$ and deferred re-weighting schedule; Balanced Softmax (Ren et al., 2020)

implements frequency-based reweighting with temperature parameter $T = 1.0$. All experiments employ 5-fold cross-validation.”

Revised text: “We additionally evaluate three recent imbalance-learning baselines using tabular architectures. DeepSMOTE uses an encoder–decoder with latent dimension 128 and encoder widths 256–128–64; minority samples are interpolated among latent-space nearest neighbors and decoded before classification, with reconstruction penalty weight 0.1. LDAM uses the same two-hidden-layer tabular MLP backbone as the other neural baselines, class-dependent margins with maximum margin $\Delta = 0.5$, scale 30, and deferred class reweighting after epoch 160. Balanced Softmax uses the same MLP backbone and adds the logarithm of the training class counts to the logits with temperature $T = 1.0$. All three baselines use Adam (learning rate 0.001, weight decay 1×10^{-5}), batch size 32, at most 200 epochs, early stopping patience 10, and stratified 5-fold cross-validation. Early stopping, learning-rate scheduling, and checkpoint selection use a held-out selection split inside each training fold, so the reported fold is evaluated only once at the end. The released code reports the seed and each fold’s Accuracy, AUC, and F1-score.” Table 7 is replaced by Revised Table 7 above (changed values in red, best-per-column value in bold), and the comparison paragraphs in Sec. 5.3 are updated accordingly in red.

Original text: “As shown in Table 7, DBUPLDM surpasses MLP-Focal in accuracy on 14 of 17 datasets, and likewise leads in AUC (13/17) and F1-score (14/17). While MLP with Focal Loss shows competitive performance on a few datasets (e.g., WDBC, Ionosphere), DBUPLDM consistently provides more reliable results, particularly on datasets with higher class imbalance and smaller sample sizes. Compared with recent deep learning methods (DeepSMOTE, LDAM, Balanced Softmax), DBUPLDM demonstrates competitive performance across all 17 evaluated datasets. While deep learning methods show strong performance on certain datasets (e.g., LDAM on Ionosphere and Australia), DBUPLDM achieves the best average performance across all three metrics (Accuracy: 84.16%, AUC: 82.78%, F1: 0.81), demonstrating its robustness and consistency. This stems from the synergistic effect of dual-balanced distance and unified pinball loss, which jointly address both class imbalance and noise sensitivity without requiring extensive architectural engineering or large-scale training data. These results validate the generalization ability and imbalance-awareness of our proposed DBUPLDM model.”

Revised text: “As shown in Table 7, compared with the MLP+Focal baseline in the new experiment, DBUPLDM achieves higher accuracy on 10 of 17 datasets and a higher average accuracy (84.16% versus 73.38%) and average F1-score (0.81 versus 0.73), whereas MLP+Focal attains higher AUC on 13 of 17 datasets and a higher average AUC (83.98% versus 82.78%). Against the three recent deep baselines, DBUPLDM remains competitive but does not dominate on every metric: DBUPLDM reaches the highest average accuracy (84.16%), Balanced Softmax has the highest average AUC (84.16%), LDAM has the highest average F1-score (0.82), and DBUPLDM reports 84.16%, 82.78%, and 0.81, respectively. We therefore do not claim that DBUPLDM is best on all three averages. Its advantages are (i) the best average classification accuracy, improving on the plain MLP+Focal baseline, (ii) high interpretability, since the decision boundary is determined by a small number of support vectors, and (iii) much lower computational cost than deep learning. DBUPLDM leads MLP+Focal on datasets such as Pima, Echo, Haberman, Statlog Heart, German, Australian, Votes, Diabetes, Fertility, and Plrx, and is competitive on the remaining benchmarks. These results validate the generalization ability and imbalance-awareness of our proposed DBUPLDM model.”

Reviewer 8, Major Concern 3: *The Wilcoxon p-value reporting needs more detail. Reporting $p = 0.03125$ for all four comparisons (DBUPLDM vs SVM, UPSVM, PinSVM, LDM) is consistent with a test over 5 or 6 paired observations where DBUPLDM wins all of them, but the paper does not say what the paired units are (datasets? η values? $\eta \times$ datasets?) or what n is. Please state the test setup and report effect sizes alongside the p-values. A one-line description would clear this up.*

Author response: We thank the reviewer for asking for the test setup: the identical p-values are expected, not an error. Each of the four Wilcoxon signed-rank tests pairs the per- η win counts of DBUPLDM against one baseline (SVM, UPSVM, PinSVM, or LDM) across the six η settings, so there are $n = 6$ paired observations per comparison (the WIN rows of the win/draw/loss table). DBUPLDM records more wins than each baseline at every η value, i.e., all six paired differences favor DBUPLDM in every comparison, for which the exact two-sided p-value is $2/2^6 = 0.03125$ in all four cases. The

revised manuscript now states this setup explicitly and additionally reports effect sizes: the matched-pairs rank-biserial correlation is 1.00 and the standardized effect size is $r = Z/\sqrt{n} \approx 0.90$ ($Z \approx 2.20$), both indicating a large effect.

Author action:

►► Location in the revised manuscript: Section 5, the paragraph below the win/draw/loss table in the “Experiments on UCI datasets” subsection.

Original text: “To validate the statistical significance of the observed performance differences, we conducted Wilcoxon signed-rank tests across the 17 datasets for each η value. Specifically, we compared the number of wins achieved by DBUPLDM against four baseline models: SVM, UPSVM, PinSVM, and LDM. The test results indicate statistically significant differences in performance across all comparisons (p-value = 0.03125 in each case), thereby reinforcing the robustness of DBUPLDM’s superiority beyond random fluctuations.”

Revised text: “To validate the statistical significance of the observed performance differences, we conducted two-sided Wilcoxon signed-rank tests comparing the win counts of DBUPLDM against four baseline models (SVM, UPSVM, PinSVM, and LDM). Each comparison pairs the two models’ win counts across the six η settings ($n = 6$ paired observations; see the WIN rows of the win/draw/loss table). DBUPLDM records more wins than each baseline at every η value, so all six paired differences favor DBUPLDM and the exact two-sided p-value is 0.03125 in all four comparisons. The matched-pairs rank-biserial correlation is 1.00 and the standardized effect size is $r = Z/\sqrt{n} \approx 0.90$ ($Z \approx 2.20$), both indicating a large effect that reinforces the robustness of DBUPLDM’s superiority beyond random fluctuations.”

Reviewer 8, Major Concern 4: *Text-table inconsistency. Section 5.3 says “deep learning methods show strong performance on certain datasets (e.g., LDAM on Ionosphere and Australia).” In Table 7, DBUPLDM beats LDAM on Australian (87.24 vs 84.20). Pick examples that actually match the table, or remove the specific claim.*

Author response: We agree that the Australia example did not support the stated comparison: DBUPLDM’s accuracy is 87.24%, compared with 84.93% for LDAM in the new experiment. We also revised the performance summary to reflect the new experiment: DBUPLDM has the highest average accuracy (84.16%), Balanced Softmax the highest average AUC (84.16%), and LDAM the highest average F1-score (0.82). Accordingly, the revised text no longer claims that DBUPLDM is best on all three average metrics. On Ionosphere, the new experiment gives LDAM accuracy of 93.73% and DBUPLDM accuracy of 78.85%; we retain this as an example of a dataset on which LDAM performs strongly.

Author action:

►► Location in the revised manuscript: Section 5.3, the comparison paragraph below Table 7 (the same paragraph reported in full under Major Concern 2).

Original text: “While deep learning methods show strong performance on certain datasets (e.g., LDAM on Ionosphere and Australia), DBUPLDM achieves the best average performance across all three metrics (Accuracy: 84.16%, AUC: 82.78%, F1: 0.81), demonstrating its robustness and consistency.”

Revised text: “Against the three recent deep baselines, DBUPLDM remains competitive but does not dominate on every metric: DBUPLDM reaches the highest average accuracy (84.16%), Balanced Softmax has the highest average AUC (84.16%), LDAM has the highest average F1-score (0.82), and DBUPLDM reports 84.16%, 82.78%, and 0.81, respectively. We therefore do not claim that DBUPLDM is best on all three averages.” (The full revised comparison paragraph is given under Major Concern 2.)

Reviewer 8, Major Concern 5: *Big Data Research scope is still only rhetorical. The new scalability paragraph in the conclusion discusses ADMM, mini-batch processing, and Random Fourier Features, but none of this is implemented or evaluated. The experiments are entirely on small UCI datasets (hundreds to about a thousand samples). At minimum, I would ask for one empirical demonstration on a larger-scale imbalanced dataset (for example, CreditCard Fraud with about 285k samples, KDD99, or a similar public benchmark) to support the scalability claims. Without this, the journal fit remains weak.*

Reviewer 8, Major Concern 6: *Novelty over the authors’ own prior work should be spelled out. The UP loss applied to a large-margin distribution classifier already appears in Gu et al. 2024 (cited in the paper). The dual balance factor is close in spirit to Cheng et al.’s CS-LDM (2016). The paper’s contribution is essentially the combination of these two ingredients. A short paragraph in the introduction or Section 3 stating what is new here relative to Gu et al. 2024 would strengthen the case.*

Reviewer 8, Minor Issue 1: *Title: “Dual Balanced Large Marginal Machine” should be “Large Margin Machine” (or “Large Margin Distribution Machine” to match the abstract). This has remained wrong across rounds and needs to be fixed.*

Reviewer 8, Minor Issue 2: *Dataset naming is inconsistent. Table 1 lists “Herberman,” the results tables use “Heberman,” and Table 7 uses “Habe.” The dataset is Haberman’s Survival. Pick one spelling and use it throughout.*

Reviewer 8, Minor Issue 3: *“Ecoil” should be “Ecoli” (E. coli dataset).*

Reviewer 8, Minor Issue 4: *“Germans” should likely be “German” (German Credit dataset).*

Reviewer 8, Minor Issue 5: *“Phonemes” vs “Phoneme” is inconsistent between Table 1 and Section 5.5.*

Reviewer 8, Minor Issue 6: *Table 7, WDBC row: “87.50/91.43/0/89” has a slash where a decimal should be. Should read “0.89.”*

Reviewer 8, Minor Issue 7: *Figure 6, subfigures (i), (j), (k), and (l) are all captioned “Prlx dataset.” These are clearly different datasets. Fix the captions.*

Reviewer 8, Minor Issue 8: *Figure 6(g) caption reads “Daibetes dataset.” Should be “Diabetes.”*

Reviewer 8, Minor Issue 9: *Section 5.5: “PinSVM, UPSVM, and DBUPLDM) performe well” should be “perform.”*

Reviewer 8, Minor Issue 10: *The Section 1 roadmap reads “Section 3 proposes the Dual Balanced method and DBUPLDM model. Section 5 discusses...” Section 4 (Properties of DBUPLDM) is skipped. Add it.*

Reviewer 8, Minor Issue 11: *Table 5 caption “Minimum distance with second order statistics” does not describe the content (ablation over balanced factors and hinge vs UP loss). Rewrite.*

Reviewer 8, Minor Issue 12: Remark renumbering: the new Remark 2 (Terminology Clarification) shifts former Remark 2 to Remark 3 and former Remark 3 to Remark 4. Please verify no downstream cross-references still point to the old numbers.

Reviewer 8, Minor Issue 13: The third contribution bullet in the introduction says “the quantile distance is introduced to replace the shortest distance between sample sets in traditional LDM.” Given the new Remark 2, this is imprecise (the replacement involves both the UP loss and the composite distance). Reword for consistency.

Reviewer 8, Minor Issue 14: The response letter header says “BDR-D-25-00303R3” but the opening paragraph says “R1,” and later describes this as the third round. Fix this.

Reviewer 8, Minor Issue 15: Reviewer 6’s comment about “deeper theoretical insight” is acknowledged in the response but not concretely addressed. A brief sentence on what was done, or why no change was needed, would help.

ACKNOWLEDGEMENT

Special thanks to you for the valuable and helpful comments and suggestions from editors and anonymous reviewers. The comments are helpful in improving our paper, as well as the important guiding significance to our research. We hope that the editors' and reviewers' concerns have been adequately addressed in the revised version.

Yours sincerely,

Pingping Gu, Denghao Dong, Quan Shi, Fudi Wang and Guojun Huang*